

Erin Avllazagaj

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RESEARCH INTERESTS

My broad research interests cover data-driven malware analysis and **automatic exploit generation**. I'm currently more focused in automatic exploit generation for the Linux kernel. Specifically, in my recent PhD work I have developed a tool (called *SCAVY*) to automatically find kernel objects that when overwritten by memory corruptions bugs can reach privilege escalation. My participation in various CTF competitions and my internship work at Meta have been major influences in this new research direction.

EDUCATION

University of Maryland

College Park, MD

Ph.D. student in Electrical and Computer Engineering, Advisor: [Prof. Tudor Dumitraş](#)

Aug. 2018 – Present

Bilkent University

Ankara, Turkey

B.S. degree in Computer Science

Aug. 2014 – May 2018

- Scholarly paper: “Privacy-Related Consequences of Turkish Citizen Database Leak” (**INFER’16**)
 - * Advisors: [Prof. Erman Ayday](#), [Prof. Ercüment Çiçek](#)
- Senior Project: “**Andlit**”
 - * Advisor: [Prof. İbrahim Körpeoğlu](#)
 - * The project is a social network of individual face recognition classifiers. Designed to help the visually impaired recognise their environment, the friends in front of them and the friends of their friends. The project was called a “third eye” by various Turkish media agencies [1*] [2] [3] [4] [5] including [daily newspapers](#).

PROFESSIONAL EXPERIENCE

[Meta \(Product Security Team\)](#) | *PhD SWE Intern* | *California, USA*

June 2022 – September 2022

- I worked with my manager Dr. David Molnar and my peers to migrate the an old Zoncolan static analysis pipeline to a new pipeline for the Privacy team to use. During this time I learned more about big data management tools like Dataswarm, Hive, Spark etc.

[Maryland Cybersecurity Center](#) | *Research Intern* | *Maryland, USA*

June 2017 – August 2017

- I worked with Octavian Suciuc to investigate code reuse of exploits in the wild. In this work I read many papers and used their static analysis techniques to measure closeness of code snippets parsed with Joern.

[4S \(contractor for HavelSan\)](#) | *Red Teaming* | *Ankara, Turkey*

July 2016 – August 2016

- I was tasked with testing a Unity based game for security issues. The game was a red-blue team simulation. The testing was important due to the client-server architecture and the security implications the server would impose at HavelSan. Without access to the source code I used tools from DevXUnityUnpacker, to Cheat Engine and IDA Pro.

PUBLICATIONS

Conferences

E. Avllazagaj, Z. Zhu, L. Bilge, D. Balzarotti, T. Dumitraş, “When Malware Changed Its Mind: An Empirical Study of Variable Program Behaviors in the Real World”, In 30th USENIX Security Symposium (USENIX Security 21), 2021

* **1st place winner** of CSAW21’ Applied Research competition [[certificate](#)]

* Press highlights: [[CSO-DE](#)] [[Cyberwire](#)] [[CSO-EN](#)] [[UMIACS](#)]

Workshops

E. Avllazagaj, E. Ayday, E. Çiçek, “Privacy-Related Consequences of Turkish Citizen Database Leak”, In International Workshop on Inference & Privacy in a Hyperconnected World (INFER’ 16), 2016

N. Garg, I. Shahid, **E. Avllazagaj**, J. Hill, J. Han, N. Roy, “Thermware: Toward side-channel defense for tiny iot devices”, The 24th International Workshop on Mobile Computing Systems and Applications (HotMobile’ 23), 2023

Posters

O. Suciuc, **E. Avllazagaj**, T. Dumitraş, “The Secret Life of Pwns: Characterizing and Predicting Exploit Weaponization”, In Conference on Applied Machine Learning for Information Security (CAMLIS’ 19), 2019

HONORS AND AWARDS

Fellowships

Clark Doctoral Fellowship GRA/GTA support (4 years)	2018 – 2022
Clark Doctoral Fellowship stipend (first 2 semesters)	2018 – 2019
Clark Doctoral Fellowship travel grant (4 years)	2018 – 2022
Bilkent University Full Tuition+Accommodation scholarship (all 4 years)	2014 – 2018
Mehmet Akif High School Tuition scholarship (all 3 years)	2011 – 2014

Awards

1 st place in CSAW21' Applied Research Competition [certificate]	2021
2 nd place in Informatics Olympiad of Tirana	2013

SERVICES

Program Committee

[Full PC Member] CSAW 22' 23'	2022, 2023
[Artifact Evaluation] Usenix Security	2021 – 2023
[Artifact Evaluation] PoPETS (Accepted papers from PETS)	2021 – 2023
[Journal Reviewer] Computer Science Review Journal	2021
[Journal Reviewer] IEEE ISCC	2022

External reviewer

Network and Distributed System Security Symposium (NDSS)	2020
USENIX Security Symposium (USENIX)	2019, 2023
International Symposium on Research in Attacks, Intrusions and Defenses (RAID)	2018, 2019
ACM Conference on Computer and Communications Security (CCS)	2019, 2020
IEEE Security and Privacy (S&P)	2022, 2023

PROJECTS

Analysis of malware behavior in the wild

Paper status

Accepted to USENIX Security 2021

Abstract: Behavioral program analysis is widely used for understanding malware behavior, for creating rule-based detectors, and for clustering samples into malware families. However, this approach is ineffective when the behavior of individual samples changes across different executions, owing to environment sensitivity, evasive techniques or time variability. The models created and tested using such traces do not account for the broad range of behaviors observed in the wild, and may result in a false sense of security.

In this project, we conduct the first quantitative analysis of behavioral variability in malware, PUP and benign samples, using a novel dataset of 7.6M execution traces, recorded in 5.4M real hosts from 113 countries. We analyze program behaviors at multiple granularities, and we show how they change across hosts and across time. We then analyze the invariant parts of the malware behaviors, and we show how this affects the effectiveness of malware detection using a common class of behavioral rules.

Automated Discovery of Memory Corruption Targets in Linux Kernel for Privesc

Paper status

Under submission

Abstract: Kernel privilege-escalation (aka. *privesc*) exploits typically leverage memory-corruption vulnerabilities to overwrite particular memory locations. These memory corruption targets play a critical role in the exploits, as they determine which privileged resources (e.g. files, memory, and operations) the adversary may access and what privileges (e.g. read, write, and unrestricted) they may gain. We propose *SCAVY*, a framework that automatically discovers memory corruption targets for privilege escalation in the Linux kernel. The key insight behind *SCAVY* is to extend the search beyond the kernel data structures that include function pointers, or pointers to such structures, and that were studied in prior work while accounting for approximately one tenth of the kernel structures.